

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: 4.2 INTRODUCTION TO SPACE PHYSICS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Answer the driving question: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? Write your answers in the space under each prompt. Use evidence from the lessons (the Big Bang, the evolution of the universe, classifying celestial bodies, the Solar System, Kepler's Laws, gravity, telescopes and space missions). For each answer, give a claim, supporting evidence and your reasoning. Write in full sentences and use the correct science words. You may include labelled sketches if they help.

Part 1: The sky looks still, but is it?

Imagine you are standing in the Rift Valley or on the shore of Lake Victoria on a clear night. The thousands of stars, the Milky Way band, the Moon and planets such as Venus and Jupiter look permanent and unchanging. Scientists say none of this existed 13.8 billion years ago. (a) Explain why the sky looks unchanging to us even though science says it has a violent, dynamic history. (b) Describe how the Big Bang is different from an ordinary explosion inside space.

(a) Claim: The sky only looks static because cosmic changes happen over timescales far longer than a human lifetime or even recorded human history. Evidence: Scientists estimate the universe began about 13.8 billion years ago. The stars, galaxies and the rest of the observable universe developed over time from that beginning, and the Solar System, including the Sun, Earth, the Moon and the planets, formed much later, about 4.6 billion years ago. Reasoning: A person watches the sky for decades, which is a tiny fraction of billions of years, so changes in stars and galaxies are too slow to notice by eye. The universe is still expanding today even though we cannot see this by looking up. (b) The Big Bang was not an explosion of matter into empty space that already existed. It was the expansion of space itself from an infinitely hot, infinitely dense singularity, producing space, time, matter and energy. The universe has been expanding and cooling ever since.

Part 2: From the Big Bang to the structured night sky

Put these events in order from earliest to latest: first stars, first atoms of hydrogen and helium, first galaxies, our Solar System, quark-gluon plasma, our Milky Way. Then explain how the

Order: quark-gluon plasma, then first atoms of hydrogen and helium (about 380,000 years after the Big Bang), then first stars (about 200 million years after the Big Bang), then first galaxies (about 1 billion years after the Big Bang), then our Solar System (about 4.6 billion years ago). Our Milky Way is one of the galaxies that formed in this galaxy-forming era, so it belongs alongside the first galaxies, before the Solar System; the exact point at which it formed in this sequence is not more precise than that. Explanation: As the universe expanded it cooled. Matter changed from a plasma into protons and neutrons and then into the first atoms. Gravity then pulled the hydrogen and helium together into clumps, which became stars and galaxies. Over vast timescales

universe changed from a hot, structureless state into the stars and galaxies visible from Nairobi tonight. In your explanation, describe what processes caused the change, and identify the type of galaxy the Milky Way is.	this produced the structured sky. The Milky Way, the glowing band we see from the Rift Valley, is a barred spiral galaxy containing over 200 billion stars, including our Sun. The Solar System formed later from a collapsing nebula. Because the light from distant stars left them millions or billions of years ago, the sky we see is a snapshot of the past, not a fixed picture.
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Part 3: Order in the Solar System and the force behind it

The planets Venus and Jupiter that we see above Nairobi belong to the Solar System. (a) Describe the layout of the Solar System, and explain what a scaled model of it (such as the corridor or toilet-paper model you walked) showed you about the space between the planets. (b) State Kepler's three laws in your own words. Then, using the orbital data you worked with for Mercury, Venus, Earth, Mars, Jupiter and Saturn, explain what you found when you compared the square of each planet's orbital period with the cube of its semi-major axis, and say what this told you about the planets. (c) Name the force that keeps the planets in orbit, state how it depends on mass and distance, and give one effect it has on Kenyan communities.	(a) The Solar System has eight planets in two zones separated by the asteroid belt, and distances are measured in astronomical units. In the scaled model the four inner planets were bunched close together near the Sun, while the outer planets, such as Neptune, were spread far apart (in our model the inner planets fitted within the first 3 squares of toilet paper and Neptune needed about 60 squares). This showed that the Solar System is overwhelmingly empty space. (b) First Law: planets orbit the Sun in ellipses with the Sun at one focus. Second Law: a planet sweeps out equal areas in equal times, so it moves faster when closer to the Sun and slower when farther away. Third Law: the square of a planet's orbital period is proportional to the cube of its semi-major axis ($T^2 \propto r^3$). When we calculated T^2/r^3 for the six planets, using T in years and r in astronomical units, the value was nearly constant, about 1 for each planet. This shows the Solar System is mathematically ordered, not random, and that planets farther from the Sun take longer to complete one orbit. (c) The force is gravity, described by Newton's Law of Universal Gravitation. The gravitational force is proportional to the product of the two masses and inversely proportional to the square of the distance between them, so doubling the distance makes the force four times smaller. The Sun's gravity, acting on each planet in this way, keeps the planets in orbit rather than flying off. The same law applies to the Moon and Earth: the Moon's gravity pulls Earth's water into tidal bulges, producing two high tides and two low tides each day at Mombasa and detectable micro-tides on Lake Victoria, which affects Kenyan port and fishing communities.
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Part 4: Gathering evidence without going there

No human was present at the Big Bang and we cannot travel to distant galaxies. (a) Name at least three kinds of evidence or instruments scientists use and	(a) Cosmic Microwave Background radiation is evidence that supports the Big Bang model. The observed expansion of galaxies shows that the universe has been expanding since it began. Telescopes gather electromagnetic radiation such as visible light, infrared and radio waves. Refracting telescopes use convex lenses to bend and focus light. Reflecting telescopes use concave mirrors and can be built much larger. Radio telescopes use large dish antennas to detect radio waves from galaxies and hydrogen clouds. Space missions such as Sputnik 1, the Hubble Space Telescope and the James Webb Space Telescope (JWST) provided
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explain what each one tells us. (b) Explain why a telescope can be described as a time machine. (c) Give one example of how Kenya or Africa contributes to space science.	further instrument-based evidence. (b) Light takes time to travel, so when a telescope collects light from a distant object it shows the object as it was when the light left it. The JWST can observe galaxies as they were only 300 million years after the Big Bang, so it looks back in time. (c) Kenya's first satellite, 1KUNS-PF, was a small CubeSat deployed from the International Space Station in 2018, and the Kenya Space Agency works in this field. Africa also takes part through the Square Kilometre Array (SKA) radio telescope project and the MeerKAT telescope in South Africa.
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Part 5: Your final answer to the driving question

Write a final explanation of 6 to 8 sentences that answers the driving question: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? Use the Claim, Evidence, Reasoning structure. Include at least one idea about the universe's origin, one about the Solar System, and one about how evidence is gathered. Then say how your thinking has changed since Lesson 1.	Claim: Everything we see in the night sky above Nairobi came from the Big Bang about 13.8 billion years ago and has evolved ever since, and scientists know this by reading the light and radiation the universe sends to us. Evidence: Cosmic Microwave Background radiation and the expansion of galaxies support the Big Bang. As the universe cooled, gravity gathered hydrogen and helium into stars and galaxies, and our Solar System formed about 4.6 billion years ago, with the Sun's gravity holding the planets in orbit. Telescopes such as the JWST collect light from objects too far away to visit. Reasoning: Because light takes time to travel, telescopes show us the past, so we can study the history of the universe without travelling to it. Change in thinking: In Lesson 1 I thought the sky was permanent and unchanging; now I understand it is a snapshot of a universe that had a beginning and is still expanding.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of the origin and evolution of the universe	Correctly describes the Big Bang as expansion of space from a singularity about 13.8 billion years ago, orders the cosmic events correctly and uses the given times accurately.	Describes the Big Bang and most events in the correct order, with minor errors in times or terms.	Describes the Big Bang as an ordinary explosion in space or places several events in the wrong order.
Use of evidence (CMB, expansion, Kepler's Laws, gravity, telescopes and missions)	Uses several specific pieces of evidence from the unit and explains clearly what each one shows.	Uses some evidence from the unit but explains what it shows only partly.	Gives few or no pieces of evidence, or only states facts without linking them to a claim.
Reasoning about indirect observation	Clearly explains that light travel time lets telescopes look back in time and that instruments extend human senses, and links this to the driving question.	Explains that instruments gather evidence from far away but is unclear about looking back in time.	Does not explain how scientists learn about places they cannot visit.

Connections to the anchoring phenomenon and Kenya	Repeatedly links ideas to the Nairobi night sky and gives accurate Kenyan or African examples such as Mombasa tides, Lake Victoria, 1KUNS-PF or the SKA.	Links to the Nairobi night sky and gives at least one accurate Kenyan or African example.	Gives little connection to the phenomenon or Kenyan examples.
Communication and Claim-Evidence-Reasoning structure	Writes a clear, well-organised answer with a claim, evidence and reasoning, using correct science vocabulary and a clear reflection on how thinking has changed.	Writes a mostly clear answer with a claim and some evidence and reasoning, and a short reflection.	Writes an unclear or incomplete answer that lacks a claim or reasoning.

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